PREWAR PIONEERS (1908–14)

In the pioneering years of flight, wealthy engineers, sportsmen, and entrepreneurs vied with one another to come up with successful designs. The basis for a nascent aircraft industry lay in the craft skills of men who made furniture, or bicycles, using fabric, wood, and piano wire. Engines evolved from those designed for early automobiles and motorcycles. The Wright Type A was not as influential as might have been expected. Wheels were generally preferred to skids, and ailerons mostly won out over the Wright's wing warping, for lateral control. The Blériot XI popularized another important configuration: the monoplane. But monoplanes were not considered sufficiently robust and suitable only for daredevil sportsmen. The aircraft configuration that predominated in WWI, the tractor biplane, emerged only slowly to predominance. This configuration began in 1909 with the Breguet and evolved into the more robust, high-performance Sopwith Tabloid.

CROSS-CHANNEL BLÉRIOT
A Blériot XI, like that used by Louis Blériot to cross the English Channel in 1909, flying over the pier in Nice, France.

Antoinette IV

The elegant fuselage and advanced, lightweight engine showed the unconventional thinking and artistic background of the Antoinette's designer, Léon Levavasseur. On July 19, 1909, Hubert Latham made the first, unsuccessful attempt to fly the English Channel in this striking French monoplane, named after its sponsor's daughter.

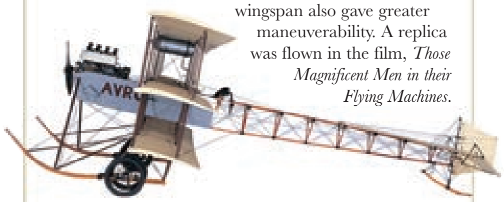
Engine 50hp fuel-injected Antoinette

Wingspan 40ft (12.2m)	**Length** 37ft (11.3m)
Top speed 45mph (72kph)	**Crew** 1

Passengers None

Avro Roe IV Triplane

In 1910, the British pioneer Alliot Verdon Roe created Avro, and this Avro IV, introduced in 1911, was the first successful triplane. The main advantage of triplanes over biplanes was that they could be built with a shorter wingspan to achieve the same lifting power. A shorter wingspan also gave greater maneuverability. A replica was flown in the film, *Those Magnificent Men in their Flying Machines.*



Engine 35hp Green 4-cylinder water-cooled

Wingspan 32ft (9.8m)	**Length** 30ft (9.2m)
Top speed Unknown	**Crew** 1

Passengers None

Breguet Tractor Biplane

After an abortive attempt in 1907 to build a helicopter, Breguet built his first biplane in 1909, using steel tubing for its main structure. The configuration of the improved 1910 version would establish the sleek, "modern" design for all future tractor-engined biplanes.

Engine 50hp 8 Renault V8 cylinder

Wingspan 42ft 9in (13m)	**Length** 25ft 9in (10m)
Top speed 44mph (70kph)	**Crew** 1

Passengers None

Bristol Boxkite

The Bristol Aeroplane Company's first successful product, the Standard Biplane, universally known as the Boxkite, was an improved version of a Henri Farman design. A total of 76 were built for military and civilian training schools around the world.

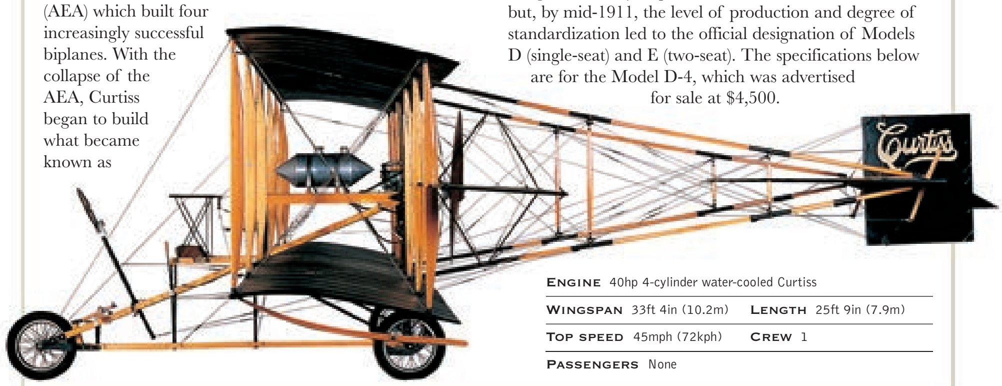
Engine 50hp Gnome rotary

Wingspan 34ft 6in (10.5m)	**Length** 38ft 6in (11.7m)
Top speed 40mph (64kph)	**Crew** 1

Passengers None

Curtiss Model D

Prior to March 1909, Glenn H. Curtiss was part of Alexander Graham Bell's Aerial Experiment Association (AEA) which built four increasingly successful biplanes. With the collapse of the AEA, Curtiss began to build what became known as the Curtiss D "Headless Pusher," based on the final AEA design. The early airplanes were all custom-built to order but, by mid-1911, the level of production and degree of standardization led to the official designation of Models D (single-seat) and E (two-seat). The specifications below are for the Model D-4, which was advertised for sale at $4,500.



Engine 40hp 4-cylinder water-cooled Curtiss

Wingspan 33ft 4in (10.2m)	**Length** 25ft 9in (7.9m)
Top speed 45mph (72kph)	**Crew** 1

Passengers None